

Unit 18: The Mean Value Theorem — Full Solutions

Warm-up

1. $f(x) = x^2 - 4x$ on $[0, 4]$

f is a polynomial, so it's continuous and differentiable everywhere — MVT applies.

$f(0) = 0$. $f(4) = 16 - 16 = 0$. Average rate of change: $\frac{0-0}{4-0} = 0$.

$f'(x) = 2x - 4$. Set $2x - 4 = 0 \Rightarrow x = 2$.

$c = 2$, which lies in $(0, 4)$.

2. $f(x) = x^2 + 2x$ on $[-1, 3]$

$f(-1) = 1 - 2 = -1$. $f(3) = 9 + 6 = 15$. Average rate: $\frac{15 - (-1)}{3 - (-1)} = \frac{16}{4} = 4$.

$f'(x) = 2x + 2$. Set $2x + 2 = 4 \Rightarrow x = 1$.

$c = 1$, which lies in $(-1, 3)$.

3. $f(x) = x^3$ on $[0, 2]$

$f(0) = 0$. $f(2) = 8$. Average rate: $\frac{8-0}{2-0} = 4$.

$f'(x) = 3x^2$. Set $3x^2 = 4 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$.

$c = \frac{2\sqrt{3}}{3} \approx 1.155$, which lies in $(0, 2)$.

4. $f(x) = x^2 - 1$ on $[1, 3]$

$f(1) = 0$. $f(3) = 8$. Average rate: $\frac{8-0}{3-1} = 4$.

$f'(x) = 2x$. Set $2x = 4 \Rightarrow x = 2$.

$c = 2$, which lies in $(1, 3)$.

5. $f(x) = x^2 - 4x + 3$ on $[1, 3]$

$f(1) = 1 - 4 + 3 = 0$. $f(3) = 9 - 12 + 3 = 0$. Since $f(1) = f(3)$, this is a Rolle's Theorem scenario.

$f'(x) = 2x - 4$. Set $2x - 4 = 0 \Rightarrow x = 2$.

$c = 2$, which lies in $(1, 3)$.

6. $f(x) = 2x^2 - 3x$ on $[0, 2]$

$f(0) = 0$. $f(2) = 8 - 6 = 2$. Average rate: $\frac{2-0}{2-0} = 1$.

$f'(x) = 4x - 3$. Set $4x - 3 = 1 \Rightarrow 4x = 4 \Rightarrow x = 1$.

$c = 1$, which lies in $(0, 2)$.

Standard

7. $f(x) = x^3 - 3x$ on $[-1, 2]$

$f(-1) = -1 + 3 = 2$. $f(2) = 8 - 6 = 2$. Since $f(-1) = f(2)$, this is a Rolle's scenario.

$f'(x) = 3x^2 - 3$. Set $3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Check the interval $(-1, 2)$: $x = 1$ lies inside; $x = -1$ is the endpoint itself, not inside the open interval, so it's discarded.

$c = 1$ (only).

8. $f(x) = \frac{1}{x}$ on $[1, 3]$

Since 0 is not in $[1, 3]$, f is continuous and differentiable throughout — MVT applies.

$f(1) = 1$. $f(3) = \frac{1}{3}$. Average rate: $\frac{\frac{1}{3} - 1}{3 - 1} = \frac{-\frac{2}{3}}{2} = -\frac{1}{3}$.

$f'(x) = -\frac{1}{x^2}$. Set $-\frac{1}{x^2} = -\frac{1}{3} \Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3}$ (taking the positive root, since we're working in $[1, 3]$).

$c = \sqrt{3} \approx 1.732$, which lies in $(1, 3)$.

9. $f(x) = x^3 - x$ on $[-1, 1]$

$f(-1) = -1 + 1 = 0$. $f(1) = 1 - 1 = 0$. Rolle's scenario.

$f'(x) = 3x^2 - 1$. Set $3x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}} = \pm \frac{\sqrt{3}}{3}$.

Both values lie inside $(-1, 1)$.

$c = \frac{\sqrt{3}}{3}$ and $c = -\frac{\sqrt{3}}{3}$ (both valid).

10. $f(x) = \sqrt{x}$ on $[0, 4]$

f is continuous on $[0, 4]$ and differentiable on the open interval $(0, 4)$ (the derivative is undefined only at the endpoint $x = 0$, which doesn't affect the open-interval requirement) — MVT applies.

$f(0) = 0$. $f(4) = 2$. Average rate: $\frac{2 - 0}{4 - 0} = \frac{1}{2}$.

$f'(x) = \frac{1}{2\sqrt{x}}$. Set $\frac{1}{2\sqrt{x}} = \frac{1}{2} \Rightarrow 2\sqrt{x} = 2 \Rightarrow \sqrt{x} = 1 \Rightarrow x = 1$.

$c = 1$, which lies in $(0, 4)$.

11. $f(x) = \sin x$ on $[0, \pi]$

$f(0) = 0$. $f(\pi) = 0$. Rolle's scenario.

$f'(x) = \cos x$. Set $\cos x = 0$. Within $(0, \pi)$, the only solution is $x = \frac{\pi}{2}$.

$c = \frac{\pi}{2}$.

12. $f(x) = x^2 - 6x + 5$ on $[0, 6]$

$f(0) = 5$. $f(6) = 36 - 36 + 5 = 5$. Rolle's scenario.

$f'(x) = 2x - 6$. Set $2x - 6 = 0 \Rightarrow x = 3$.

$c = 3$, which lies in $(0, 6)$.

Challenge

13. $f(x) = \frac{1}{x}$ on $[-1, 1]$

MVT does not apply. The function $f(x) = \frac{1}{x}$ is not continuous at $x = 0$ (it's not even defined there — division by zero), and $x = 0$ lies right inside the interval $[-1, 1]$. Since continuity on the entire closed interval is a required hypothesis, and it fails at this interior point, we cannot use MVT here at all — there's no guarantee that a value c with the required property exists.

14. $f(x) = |x|$ on $[-2, 2]$

MVT does not apply. The function $f(x) = |x|$ is continuous everywhere (no breaks), but it is not differentiable at $x = 0$ — it has a sharp corner there, as we established back in Unit 7. Since $x = 0$ lies inside the open interval $(-2, 2)$, and differentiability on the entire open interval is a required hypothesis, this hypothesis fails, so MVT's guarantee does not apply.

15. $f(x) = x^3 - 3x^2 + 2$ on $[0, 3]$

$f(0) = 2$. $f(3) = 27 - 27 + 2 = 2$. Rolle's scenario.

$f'(x) = 3x^2 - 6x = 3x(x - 2)$. Set $3x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$.

Check the open interval $(0, 3)$: $x = 0$ is the endpoint itself, so it's discarded. $x = 2$ lies inside $(0, 3)$.

$c = 2$ (only).

16. Suppose $f'(x) \leq 2$ everywhere, with $f(0) = 0$ and $f(3) = 10$.

By the Mean Value Theorem (assuming f is differentiable everywhere, which makes it automatically continuous too), there must exist some c in $(0, 3)$ such that:

$$f'(c) = \frac{f(3) - f(0)}{3 - 0} = \frac{10 - 0}{3} = \frac{10}{3} \approx 3.33$$

But we assumed $f'(x) \leq 2$ for every value of x , including $x = c$. This means we'd need $f'(c) \leq 2$ and $f'(c) = \frac{10}{3}$ at the same time — but $\frac{10}{3} > 2$, which is a direct contradiction.

Conclusion: no such function can exist. A function whose derivative never exceeds 2 anywhere simply cannot climb from $f(0) = 0$ up to $f(3) = 10$ — that would require an average rate of change of $\frac{10}{3}$, which is faster than the derivative is ever allowed to be.

17. f continuous on $[1, 5]$, differentiable on $(1, 5)$, $f(1) = 3$, $f(5) = 15$.

Average rate of change: $\frac{15 - 3}{5 - 1} = \frac{12}{4} = 3$.

By MVT, there exists some c in $(1, 5)$ such that $f'(c) = 3$.

In words: if $f'(x)$ stayed strictly below 3 for the entire interval, the function couldn't possibly climb fast enough on average to go from 3 up to 15 — its overall average rate would necessarily be less than 3, contradicting the actual computed average of exactly 3. Likewise, if $f'(x)$ stayed strictly above 3 the whole time, the function would climb too fast on average, again contradicting the true average. So the derivative is forced to actually equal 3 at some point in between — it can't just hover above or below that value forever.

Applied

18. A car travels 150 miles in 3 hours.

$$\text{Average velocity} = \frac{150}{3} = 50 \text{ mph.}$$

Let $s(t)$ be the car's position, a continuous and differentiable function of time (a reasonable physical assumption — a car's position doesn't teleport or jump). By the Mean Value Theorem, there must exist some time c during the 3-hour trip where

$$s'(c) = 50 \text{ mph}$$

Since $s'(t)$ represents the car's instantaneous velocity (what the speedometer reads), this guarantees the speedometer read exactly 50 mph at some instant during the drive — no matter how the car sped up, slowed down, or varied its speed throughout the trip.

19. Profit $P(0) = 200$, $P(12) = 1400$.

$$\text{Average rate of change: } \frac{1400 - 200}{12 - 0} = \frac{1200}{12} = 100 \text{ dollars/month.}$$

By MVT, there exists some c in $(0, 12)$ such that $P'(c) = 100$.

This guarantees there was some specific moment during the year when the company's profit was growing at an instantaneous rate of exactly \$100 per month.

20. A 10 km race completed in 40 minutes.

$$\text{Convert 40 minutes to hours: } 40 \div 60 = \frac{2}{3} \text{ hour.}$$

$$\text{Average speed: } \frac{10}{2/3} = 10 \times \frac{3}{2} = 15 \text{ km/h.}$$

By MVT (assuming the runner's position is a continuous, differentiable function of time — a reasonable assumption for a real race), there must exist some instant during the race where the runner's instantaneous speed was exactly 15 km/h.

21. $h(0) = 0$, $h(10) = 2000$.

$$\text{Average velocity: } \frac{2000 - 0}{10 - 0} = 200 \text{ ft/s.}$$

By MVT, there exists some c in $(0, 10)$ such that $h'(c) = 200$.

This means at some moment during the 10-second flight, the rocket's instantaneous velocity was exactly 200 ft/s — it couldn't have stayed entirely above or entirely below that value the whole time and still average out to 200 ft/s overall.